Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/02/2024
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Optimizing the Combination of Intranasal Scopolamine and Sensory Augmentation to Mitigate G-Transition Induced Motion Sickness and Enhance Sensorimotor Performance		
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Program/Discipline--Element/Subdiscipline:			
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Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Daniels, Vernie M.S. (KBR/NASA Johnson Space Center) Chough, Natacha M.D. (NASA Johnson Space Center)		
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Task Description:	Motion sickness represents one of the greatest clinical challenges impacting crew activities during and following g-transitions. Shorter duration missions involving new commercial vehicles and/or Artemis lunar missions will require continued emphasis on motion sickness countermeasures for early inflight prevention and treatment. The higher incidence of re-entry motion sickness following longer duration spaceflights coupled with the challenges associated with capsule egress during water landings also impose greater risks for crew recovery operations. Our overall goal is to characterize the effectiveness of motion sickness countermeasures to improve inflight and postflight recovery for future space travelers on these various platforms. We are conducting both controlled laboratory experiments of specific countermeasures during capsule wave motion simulation and conducting field testing in operational environments to characterize the incidence of motion sickness during various mission phases, and the efficacy of motion sickness countermeasures, if any, are utilized. The aims of our laboratory studies include evaluation of intranasal scopolamine and sensory augmentation to mitigate motion sickness and enhance crew performance. The intranasal form of scopolamine has the advantage of rapid bioavailability (i.e., therapeutic plasma concentration) with minimal side effects. This formulation allows crewmembers to self-medicate in a suited environment either before or after the onset of symptoms. Water landings may involve provocative wave motion during which crewmembers are deprived of a stable Earth reference inside the crew capsule. Sensory augmentation, e.g., vibrotactile feedback of Earth vertical, has been effective as a spatial awareness and balance aid with vestibular impairment. We hypothesize that both intranasal scopolamine and sensory augmentation of Earth vertical, either administered separately or combined, will be effective to mitigate motion sickness and improve task performance. The initial pilot ground study involved validation of a wave motion stressor to induce sickness, and evaluation of sensory augmentation in this simulated wave motion environment. A multi-degree of freedom platform with the subject seated in an enclosed cabin mockup was utilized to simulate the provocative capsule motion during water landings. Performance on a series of functional tasks (tilt motion tracking with and without a paced auditory serial addition test (PASAT) dual-task, eye-head-hand target acquisition, psychomotor vigilance test) was measured pre, during, after capsule wave motion. The capsule motion consisted of three 15 min periods of combined pseudorandom pitch, roll and heave that continued until the subject reached a motion sickness endpoint representing severe malaise on the Pensacola Diagnostic Index (up to 45 min maximum duration). The first study aim (1a) focused on prevention of motion sickness using intranasal scopolamine using a double-blinded repeated measures design in 30 subjects. Intranasal scopolamine was provided by Defender Pharmaceuticals, Inc. (DPI-386 Nasal Gel, referred to as Inscop) self-administered by a nasal pump (Aptar Pharma) that delivers 0.4 mg dose (0.2 mg / nostril). Motion sickness symptom onset, severity, and recovery were compared across treatment and placebo control sessions counterbalanced across subjects and separated by at least one week. The bioavailability of scopolamine for each session will be estimated from plasma concentrations obtained every 15 min. Cognition (psychomotor vigilance task) and subjective reports of drug side effects were obtained. Based on the pilot study, operational performance was assessed during the capsule wave motion using tilt motion tracking and a tablet-based eye-hand target acquisition task. The second part of this laboratory aim (1b) will be to evaluate sensory augmentation with and without intranasal scopolamine to prevent motion sickness during simulated capsule wave motion. For specific aim 2, a laboratory-based study will be used to evaluate the efficacy of intranasal scopolamine to provide treatment (“rescue”) of symptoms following motion sickness onset during simulated capsule wave motion. We are currently evaluating utilizing the advanced capsule wave motion capabilities of the Disorientation Research Device (aka Kraken) at the Naval Medical Research Unit – Dayton for aim 1B and/or aim 2. Specific aim 3 will evaluate the feasibility and efficacy of administering the intranasal scopolamine in operational field settings using both astronaut and ground-control subjects that are exposed to provocative motion as part of their assigned duties. For the ground-control subjects, these may involve a number of operational environments including motion simulations (e.g., high-g profile simulations during centrifugation), parabolic flights and/or Orion capsule recovery operations at sea. For the astronaut participants, we are recruiting from free-flier missions (e.g., Polaris Dawn), and both Private Astronaut Missions (PAM) and United States Orbital Segment (USOS) crewmembers assigned to the missions to the International Space Station (ISS). Astronaut participants may choose to test Inscop during provocative preflight training exercises (e.g., centrifugation), and can choose to take the medication prophylactically to prevent symptoms or after symptom onset to treat motion sickness during the launch and/or landing mission phases. Both ground-control and astronaut participants will be required to test the medication during a training session to monitor for adverse side effects. Participants will complete a short debrief questionnaire to capture motion sickness symptoms, side effects, and feasibility comments each time they take the medication. We will also include astronaut “control” subjects who do not take Inscop to comment on motion sickness severity within the initial early inflight and postflight periods, what countermeasures they did use and rate their effectiveness. While this study aim is not blinded, the inclusion of both active and control subjects will provide a more complete characterization of the impacts of motion sickness on crew activities during and following g-transitions, and the effectiveness of motion sickness countermeasures to improve inflight and postflight recovery. In addition, we are conducting a retrospective review of medical records from both the Shuttle and ISS programs to include a more comprehensive characterization of the motion sickness risks during missions with different vehicles and mission durations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Intranasal scopolamine provides crewmembers with the ability to self-administer medication for prevention and/or treatment of motion sickness during critical mission phases, including launch, landing, and recovery operations. The rapid bioavailability, minimal side effects, and ability to self-administer real-time dosage adjustments make this an ideal formulation for other operational environments that involve provocative motion, e.g., military pararescue and emergency medicine, as well as entertainment platforms, e.g., boating and virtual reality. Sensory augmentation using vibrotactile feedback of body position has shown promise as an effective rehabilitation tool for vestibular disorders and piloting aid. The combination of non-pharmaceutical approaches like sensory augmentation with intranasal scopolamine has the benefit to mitigate motion sickness and enhance crew performance over a variety of spaceflight and earth-based motion platforms.

Task Progress:	October 2025 Update: Per NASA's Human Research Program, this investigation has been discontinued. The reporting below is the final progress report (Ed., 10/3/25). Aim 1a Lab Study: The results from the laboratory study evaluating intranasal scopolamine using a double-blinded repeated measures design were reported in the 2023 annual report and presented at the NASA Human Research Program (HRP) Investigators' Workshop (Wood et al., 2024). [Ed. Note: See Bibliography.] While our results were consistent with previous findings that intranasal delivery of scopolamine can be effective at reducing motion sickness symptoms, efficacy was only demonstrated after >45 min from self-administration with the current gel formulation (Defender Pharmaceuticals, Inc., DPI-386 Nasal Gel). The plasma concentration reached near-peak levels by the start of the testing and remained elevated throughout the 45 min motion sickness testing; however, the concentrations were quite variable across subjects. The change in time-to-motion sickness endpoint between placebo and treatment sessions was moderately correlated with plasma concentration (rho =0.48, p = 0.056), improving on average 4.4 ± 18.4 min, mean ± std with Inscop versus placebo. This variability in plasma concentration clearly presents challenges when considering this formulation for operational environments. The time-to-motion sickness endpoint was based on severe malaise, defined as symptom score of = 8 points using the Pensacola Diagnostic Index (PDI) (Graybiel et al., 1968). We also employed a Subjective Discomfort Rating (SDR) using a 0-20 point scale (Oman et al., 1986; Keshavarz and Hecht, 2011). Since the 0-20 scale has been implemented in our flight studies as a quick motion sickness scoring technique, we utilized the results of this study along with another recent HRP-funded motion sickness study (Mayo Clinic, Pradhan et al., 2024) to compare these two scales (Gleed et al., 2024). In addition to demonstrating that the SDR is highly correlated to the multi-symptom scale (rho = 0.7, p < 0.001), this analysis provides a criteria for classifying severity levels when using the SDR. For example, SDR scores less than 3 were not associated with motion sickness endpoints and SDR scores greater than 12 were typically associated with the greatest severity on the PDI scale. Aim 3 Field Testing: Aim 3 was established to assess the feasibility on intranasal scopolamine during spaceflight, and to better characterize the effectiveness of other motion sickness countermeasures (control subjects). One of the major efforts of this past year was to expand the motion sickness symptom questionnaire used during early inflight and postflight phases to include multiple symptoms while maintaining the SDR scale. This questionnaire has also been incorporated in the Spaceflight Standard Measures study as well as the CIPHER Vestibular Health study. [Ed. Note: CIPHER is the short title for a set of 14 studies sponsored by NASA and international partner agencies (https://). CIPHER stands for "Complement of Integrated Protocols for Human Exploration Research".] The version used inflight on the International Space Station (ISS) has been implemented on the European Space Agency's (ESA) EveryWear application, while the postflight questionnaires use the Qualtrics platform. This past year, questionnaires were obtained from four control subjects on the free-flyer SpaceX Polaris Dawn flight and four control subjects on Axiom- 3 Private Astronaut Mission to the ISS. Seven USOS crewmembers were recruited (6 active and 1 control). Unfortunately, the NASA Institutional Review Board (IRB) determined that an emergent situation warranted study suspension so the data collection is currently on hold. Retrospective Data Mining: In addition to our prospective study, we have been summarizing medical reports of both inflight space motion sickness and reentry motion sickness following landing, using data from both Shuttle and ISS missions. During this past reporting period, Dr. Blue conducted an exhaustive review through the electronic medical records. This new data set is being compared with that obtained from the Longitudinal Study of Astronaut Health (LSAH) data archive and will be used to evaluate different risk factors such as sex, prior flight experience, body mass index, flight duration and launch / landing vehicles.
Bibliography Type:	Description: (Last Updated: 02/20/2026)
Abstracts for Journals and Proceedings	Wood SJ, Beltran NE, Bollinger AM, Macaulay TR, Daniels VR, Wang Z. "Evaluation of intranasal scopolamine for the prevention of wave motion induced motion sickness while maintaining performance on operationally relevant tasks." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024